Topic 6 Review [84 marks]

1. [Maximum mark: 4] 24M.1.HL.TZ1.9

Outline **two** operating system resource management techniques. [4]

2. [Maximum mark: 15] 23N.1.HL.TZ0.12

Input devices that detect cars approaching a crossroads are connected to a microprocessor.

(a.i) Identify **two** types of sensor that can be used to detect approaching cars. [2]

(a.ii) Outline why sensors are appropriate input devices in this situation. [2]

(b) Suggest the type of memory that could be used to store the control program in the microprocessor. [2]

The traffic lights at the crossroads are also connected to a microprocessor. A person who wishes to cross the road presses a button at a traffic light. This causes an interrupt.

(c.i) Outline what is meant by an interrupt. [2]

(c.ii) Explain how the microprocessor can deal with this interrupt. [3]

Cameras are installed on the top of the traffic lights at the crossroads.

(d.i) Outline **one** benefit of monitoring the traffic with cameras. [2]

(d.ii) Outline **one** concern about monitoring the traffic with cameras. [2]

3. [Maximum mark: 4] 23M.1.HL.TZ1.4

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Sensors that take readings of the levels of different pollutants have been installed at a number of sites along a river. Each reading is sent to a central computer, where it is processed and analysed.

(a) Define the term *interrupt*. [1]

(b) Describe how polling could be used in this situation. [3]

4. [Maximum mark: 3] 22N.1.HL.TZ0.7

Explain how increasing the size of the central processing unit (CPU) cache improves the performance of a computer. [3]

5. [Maximum mark: 3] 21N.1.HL.TZ0.8

Explain how an operating system manages peripherals. [3]

6. [Maximum mark: 3] 22M.1.HL.TZ0.8

State **three** operating system resource management techniques. [3]

7. [Maximum mark: 1] 21M.1.HL.TZ0.9

Identify **one** advantage of using a dedicated operating system on a mobile phone. [1]

8. [Maximum mark: 18] 21M.1.HL.TZ0.15

A business has a range of different computers within the organization, including laptops, desktops and file servers. Wherever possible the organization uses a common operating system on its computers.

(a) Outline **two** resource management techniques that are likely to be carried out by the operating system of a desktop computer. [4]

(b) Outline **one** way the operating system hides the complexity of the hardware from the computer user. [2]

Memory requirements and processor speed will vary depending on the tasks required of the computer.

(c.i) Contrast the memory requirements of a laptop computer and a file server. [2]

(c.ii) Contrast the processor speed requirements of a laptop computer and a file server. [2]

The business has decided to implement a computer-based system to switch the room lights on and off automatically. The lights will only be switched on if the level of light is below a specific reading and there are people in the room. The lights will be switched off when the room has been unoccupied for at least five minutes.

(d) State **two** types of sensor that are required to control the lighting to ensure it switches on when it is required. [2]

(e) Explain how the system makes use of the data it receives from the sensors to determine when to switch the lights on. [4]

(f) Outline how the system will prevent the lights from being switched off too quickly when it thinks the room is unoccupied. [2]

9. [Maximum mark: 3] 20N.1.HL.TZ0.3

Describe how an operating system uses paging when running a program. [3]

10. [Maximum mark: 15] 20N.1.HL.TZ0.11

A company produces and sells domestic floor-cleaning robots.

The floor-cleaning robots can clean different surfaces like wood and carpet. The floor-cleaning robots can also avoid obstacles or stairs.

Sensors are used by the processor that controls the floor-cleaning robot so that it can move safely.

(a) Describe **two** types of sensors used in the floor-cleaning robots. [4]

(b) Explain the function of an output transducer in this situation. [3]

A computerized security system for the company's headquarters protects against unauthorized access using a swipe-card system. Each door has a swipe-card reader that is connected to the central computer. A database stores the IDs of all employees and the rooms they are allowed to access.

(c.i) Identify **one** alternative computerized method that could be used in place of the swipe-card readers. [1]

(c.ii) Describe how the method identified in (c)(i) functions. [3]

(d) Compare polling **and** interrupts as mechanisms for the swipe-card readers to interact with the central computer. [4]

11. [Maximum mark: 15] 19N.1.HL.TZ0.11

A washing machine manufacturer has created its website to be viewed on standard desktop computers as well as mobile devices. The mobile browsing experience differs from desktop browsing.

(a.i) Define the term *screen resolution*. [1]

(a.ii) Describe **two** issues resulting from the website being viewed on various devices, such as desktops and smartphones.

[4]

Different devices such as desktop computers and mobile devices have different operating systems.

(b) Explain the role of the operating system (OS) in terms of managing the hardware resources. [4]

(c) A washing machine uses a control system.

The microprocessor controls the washing machine and its actions. To complete the wash and rinse process the user selects the program, loads the washing machine and pushes the start button.

Describe the interaction between the sensors, microprocessors and output transducers in this situation. [6]